

Wine Quality Exploratory Data Analysis

Objective

To explore the factors affecting wine quality and identify relationships between different physicochemical properties and wine quality scores.

Dataset

Wine Quality Dataset obtained from Kaggle/UCI Repository.

Data Cleaning

- Loaded the dataset using Pandas.
- Checked for missing values.
- Removed duplicate records.
- Identified and treated outliers using boxplots and the IQR method.
- Verified data types and ensured data consistency.

Statistical Analysis

- Calculated mean, median, mode, variance, and standard deviation.
- Analyzed the distribution of wine quality scores.
- Examined the distribution of numerical features such as alcohol, acidity, and pH.

Correlation Analysis

- Generated a correlation matrix.
- Identified highly correlated features.
- Analyzed the relationship between alcohol content and wine quality.
- Examined correlations among acidity-related variables.

Visualizations Created

- Histogram of Wine Quality Scores
- Correlation Heatmap
- Boxplots for Outlier Detection
- Scatter Plot of Alcohol vs Quality
- Bar Chart of Average Quality Scores

Key Observations

1. Alcohol content showed a positive relationship with wine quality.
2. Volatile acidity was negatively correlated with wine quality.
3. Most wines were rated in the medium-quality range.
4. Several numerical features contained outliers.
5. Some chemical properties exhibited strong correlations with each other.

Conclusion

The analysis revealed that wine quality is influenced by several physicochemical properties, particularly alcohol content and acidity levels. Correlation analysis helped identify important relationships among features, while visualizations provided a clear understanding of data patterns. This project demonstrated how exploratory data analysis can be used to uncover valuable insights from real-world datasets.